

AI Placement Predictor

Decode Employability DNA: Institutional Career Readiness & Upskilling Engine

Domain: Data Science / Artificial Intelligence

Target Architecture: Full-Stack Institutional Analytics & ML Web Platform

Time Limit: 16 Hours

Team Size: Up to 4 Members

1. Background & Problem Overview

In today's competitive landscape, placement readiness depends on multidimensional factors beyond academic grades—spanning technical certifications, completed projects, practical internships, logical aptitude, and communication proficiencies. However, college placement offices often lack data-driven systems to identify which students are genuinely 'placement-ready' and who requires targeted institutional upskilling.

Participants must design and build the **AI Placement Predictor**, an internal institutional intelligence platform tailored exclusively for **students and Training & Placement Officers (TPOs)**. The system models a student's employability profile, calculates transparent placement probabilities, diagnoses specific skill deficits, auto-generates personalized career roadmaps, and equips placement officers with actionable department-wide cohort analytics—without external corporate job-application portals.

2. Core Challenge & Scope Boundary

The system must solve an **explainable predictive modeling and institutional gap-analysis problem**, not a job board or recruitment ATS. It operates entirely within the college ecosystem, providing diagnostic transparency: students learn exactly how to improve their employability standing, while placement officers pinpoint vulnerable student segments requiring departmental training interventions.

Input Dimensions for Employability Modeling

- **Academic Track Record:** CGPA, 10th/12th percentages, backlogs history.
- **Technical Competencies:** Core programming languages, frameworks, verified certifications.
- **Practical Experience:** Project complexity, open-source work, completed internships.
- **Aptitude & Problem Solving:** Quantitative, logical, and coding assessment benchmarks.
- **Soft Skills & Communication:** Verbal fluency, interview ratings, presentation scores.
- **Extracurricular Activities:** Hackathon participation, leadership roles, tech societies.

3. Mandatory System Modules & Functional Requirements

1. Student Profile & Data Ingestion Engine

A structured portal where students configure or upload their multi-tier academic and skill profile, supporting manual entry or CSV upload of academic records, technical certifications, project links, and self/assessed soft skill ratings.

2. ML-Driven Employability & Placement Probability Engine

- **Predictive Classification:** Trains an ML model (e.g., Random Forest, XGBoost, or Logistic Regression) on anonymized historical student datasets to calculate placement probability (0–100%) and readiness status (Ready, Near-Ready, Needs Training).
- **Target Role Mapping:** Predicts alignment and confidence levels across institutional career tracks (e.g., Full-Stack Developer, Data Analyst, Cloud/DevOps Engineer, QA Specialist).

3. Explainable AI (XAI) & Factor Transparency

The platform must avoid black-box numerical outputs. Utilizing XAI methodologies (such as SHAP values, LIME, or tree feature importance), the system must visualize and explain exactly which parameters boosted or penalized a student's employability score (e.g., *" +18% due to full-stack internships; -14% due to missing database certifications"*).

4. Automated Skill Gap Diagnostic & Personalized Roadmap

- **Skill Gap Analysis:** Compares the student's evaluated competencies against target career profile benchmarks to isolate missing skills.
- **Actionable Upskilling Plan:** Dynamically outputs structured, phased recommendations (e.g., required courses, specific project topics, interview drill areas) with realistic time-to-completion estimates.

5. TPO (Training & Placement Officer) Institutional Dashboard

A high-level command center for college administrators and placement officers providing:

- **Departmental Readiness Metrics:** Overall and branch-wise placement readiness percentages (CSE, ISE, ECE, Mech, etc.).
- **Cohort Vulnerability Filters:** Instant identification of students falling below readiness thresholds (<60%) to schedule targeted bootcamp training.
- **Institutional Skill Deficit Heatmaps:** Visualizing widespread skill voids across the batch (e.g., *"64% of 6th-semester students lack SQL proficiency"*).

4. 16-Hour MVP Deliverables

- A responsive web interface with dedicated **Student Diagnostics View** and **TPO Analytics Dashboard**.
- A trained and evaluated ML pipeline processing student attributes and outputting probabilistic readiness scores.
- Interactive Explainable AI visualizations (feature contributions via SHAP/LIME or bar charts).
- Dynamic skill-gap diagnosis and personalized roadmap generation for at least two career tracks.
- Departmental aggregation view showing batch-wide employability distributions and cohort filters.

Technical Implementation Hint: Do not waste time building live scraping or external corporate portals. Focus on the core ML pipeline, feature explainability, and clean dashboard visualization using Streamlit, Plotly Dash, or React+FastAPI with an anonymized or synthetic student dataset.